

THE JOINT LAW OF ZERO GAPS AND LOCAL MAXIMA OF HARDY'S Z -FUNCTION: A NUMERICAL STUDY AND A CUE EFFECTIVE-DIMENSION ANOMALY

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ABSTRACT. For consecutive ordinates $\gamma_n < \gamma_{n+1}$ of nontrivial zeros of the Riemann zeta function, let $\delta_n = \gamma_{n+1} - \gamma_n$ and let $M_n = \max_{\gamma_n \leq t \leq \gamma_{n+1}} |Z(t)|$, where Z is Hardy's function. The mean of M_n^2 is known: Conrey and Ghosh determined its leading order, and Hughes, Lugmayer and Pearce-Crump recently obtained the lower-order terms. The *joint* law of (δ_n, M_n) appears not to have been studied numerically. Using 3.4×10^5 gaps up to height $t = 1.6 \times 10^6$, we find that (i) the sample mean of M_n^2 matches the proven asymptotics to 0.05% (slope) and 1% (constant term), validating the dataset; (ii) the Pearson correlation of (unfolded) gap and maximum is $r \approx 0.75\text{--}0.89$ per window, far above the value 0.505 ± 0.048 attained by phase-randomized Gaussian surrogates with the same power spectrum ($+6.7\sigma$); (iii) compared with characteristic polynomials of Haar-random unitary matrices (CUE), the zeta correlation at height t matches CUE of *effective dimension* $N_{\text{eff}} = L + c$, $L = \log \frac{t}{2\pi}$, with $c_{\text{Pearson}} = 0.93 \pm 0.15$ and $c_{\text{Spearman}} = 1.86 \pm 0.15$, each constant over $5.6 \leq L \leq 12.45$; and (iv) since no single effective dimension fits both statistics, the joint gap–maximum law of ζ is not that of any finite-dimensional CUE. We propose the arithmetic (prime-driven) origin of the two shift constants as an open problem.

1. INTRODUCTION

Let $Z(t)$ denote Hardy's function, so that $|Z(t)| = |\zeta(\frac{1}{2} + it)|$ and the real zeros of Z are the ordinates of the critical-line zeros of ζ . Write γ_n for the n -th positive ordinate, $\delta_n = \gamma_{n+1} - \gamma_n$ for the n -th gap, and

$$M_n = \max_{\gamma_n \leq t \leq \gamma_{n+1}} |Z(t)|$$

for the height of the wave between two consecutive zeros.

The *marginal* statistics of both quantities are classical objects. The gaps, after unfolding, follow GUE spacing statistics to remarkable accuracy (Montgomery [7], Odlyzko [8]); the maxima satisfy, under the Riemann Hypothesis,

$$(1) \quad \frac{1}{N(T)} \sum_{0 < \gamma_n \leq T} M_n^2 \sim \frac{e^2 - 5}{2} \log T,$$

by Conrey and Ghosh [2], with the full descending chain of lower-order terms obtained recently by Hughes, Lugmayer and Pearce-Crump [3]; see also [9] for the first moment with derivatives, and Milinovich [6] for bounds on higher moments.

The *joint* law of the pair (δ_n, M_n) is qualitatively constrained by folklore: Lehmer pairs [5] (very close zeros) force a very low intermediate maximum, and the large-gaps methods of Conrey–Ghosh–Gonek implicitly exploit the converse. Quantitatively, however, we have not found in the literature: a measured correlation coefficient for (δ_n, M_n) ; a comparison of that

correlation against spectrum-matched Gaussian surrogates; or a comparison against the CUE characteristic-polynomial model of Keating and Snaith [4]. This note supplies these three measurements and records a structural surprise in the third one.

Our effective-dimension framing follows Bogomolny, Bohigas, Leboeuf and Monastera [1], who showed that finite-height deviations of the *spacing distribution* of zeta zeros are captured by CUE of a well-chosen finite effective dimension. The observable studied here — the gap–maximum coupling — turns out to select its own effective dimension, and different projections of the joint law select *different* ones.

2. DATA AND VALIDATION

Two independent numerical engines were used.

Engine A (to $t \leq 7.5 \times 10^4$). Zeros from a standard table; M_n computed by bounded scalar minimization of $-|Z|$ per gap using arbitrary-precision evaluation of Z (mpmath), 10^5 gaps in total.

Engine B (windows up to $t = 1.6 \times 10^6$). A vectorized Riemann–Siegel implementation (main sum plus the C_0 correction), with zeros located by dense sign-change bracketing plus bisection, close pairs recovered by a targeted fine re-scan, and M_n from a 24-point interior grid with parabolic refinement. Against arbitrary-precision values the pointwise error is at most 7.6×10^{-6} over $10^5 \leq t \leq 1.6 \times 10^6$. Six windows of 4×10^4 gaps each were computed, centered at $t = 1.2, 2.0, 3.5, 6.0, 10, 16 \times 10^5$. In the overlap region the two engines agree to 10^{-4} in the gaps and to 0.1% (median: exact) in the maxima.

Both engines were validated against proven results. Writing $L = \log \frac{t}{2\pi}$ and fitting the windowed means of M_n^2 linearly in L ,

$$\widehat{\text{mean}}(M_n^2 | t) = aL + b,$$

we find $a = 1.1939$ (after correcting a measured 0.1% optimizer bias) against the Conrey–Ghosh constant $\frac{1}{2}(e^2 - 5) = 1.19453$, a deviation of -0.05% . For the constant term, differentiating the summatory asymptotic of [3] gives the local prediction

$$\text{mean}(M_n^2 | t) = AL + (\alpha_{-1} + 2A) + \frac{\alpha_0 + \alpha_{-1}}{L} + O(L^{-2}), \quad A = \frac{e^2 - 5}{2},$$

with $\alpha_{-1} = 5 - e^2 - 10\gamma_0 + 2e^2\gamma_0 = 0.368945$ and $\alpha_0 = -0.423285$ (Stieltjes constants γ_0, γ_1). This zero-free-parameter curve passes through all 26 of our windowed means within 2σ ; the fitted $b = 2.776$ deviates from the predicted effective value 2.749 by $+0.97\%$. We regard the dataset as anchored to two proven/published results.

3. THE CORRELATION AND TWO NULL MODELS

Within each window we unfold both variables using known scales: $\tilde{\delta}_n = \delta_n \cdot L/2\pi$ and $\tilde{M}_n = M_n/\sqrt{AL + 2.758 - 0.054/L}$, so that $\text{mean}(\tilde{\delta}) = 1$ and $\text{mean}(\tilde{M}^2) = 1$ to within 10^{-3} . Table 1 reports the windowed Pearson correlation r of $(\tilde{\delta}_n, \tilde{M}_n)$.

Null 1: spectrum-matched Gaussian surrogates. Following the surrogate-data method of Theiler et al. [10], one generates Gaussian processes with the same power spectrum as $Z(t)$ (random Fourier phases), extracts their zeros and between-zero maxima, and computes the same correlation. In an earlier session of this project (over $t \in [14, 1000]$) this null gave $r_{\text{null}} = 0.505 \pm 0.048$ over 30 trials against the zeta value $r = 0.825$ on the same range: a

TABLE 1. Windowed correlation of unfolded gap and maximum, and the effective CUE dimension shift (Section 4).

L	n (gaps)	r (Pearson)	$N_{\text{eff}}^P - L$	$N_{\text{eff}}^S - L$
5.60	1 029	0.8917	+0.93	+1.83
6.30	2 322	0.8735	+0.93	+1.88
6.99	5 174	0.8566	+0.90	+1.82
7.69	11 418	0.8394	+0.92	+1.88
8.39	24 988	0.8238	+0.92	+1.87
9.08	54 299	0.8096	+0.90	+1.86
9.86	39 992	0.7947	+0.90	+1.83
10.37	39 997	0.7849	+0.94	+1.83
10.93	39 998	0.7754	+0.95	+1.82
11.47	39 998	0.7663	+1.00	+1.94
11.98	39 999	0.7589	+0.99	+1.88
12.45	40 000	0.7521	+1.01	+1.93

+6.7 σ excess. The gap–maximum coupling of ζ is therefore far stronger than that of *any* Gaussian process with its spectrum; it depends on structure beyond second-order statistics.

Null 2: CUE characteristic polynomials. For an $N \times N$ Haar-random unitary U with eigenphases ϕ_j , put $|\Lambda(\theta)| = \prod_j 2 |\sin \frac{\theta - \phi_j}{2}|$, and for each of the N cyclic eigenphase gaps record the pair (gap, interior maximum of $|\Lambda|$). The density matching $N \leftrightarrow L$ is the standard one of [4]. Table 2 reports the correlation for $N = 5, \dots, 19$ (2×10^5 gaps per N); we are not aware of a published computation of this quantity.

TABLE 2. Gap–maximum correlation for CUE characteristic polynomials.

N	Pearson	Spearman	N	Pearson	Spearman
5	0.9293	0.9684	13	0.7628	0.8862
6	0.9004	0.9541	14	0.7494	0.8795
7	0.8758	0.9423	15	0.7347	0.8741
8	0.8508	0.9303	16	0.7228	0.8672
9	0.8303	0.9198	17	0.7120	0.8609
10	0.8107	0.9102	18	0.6961	0.8562
11	0.7929	0.9010	19	0.6910	0.8512
12	0.7773	0.8934			

At the conventional matching $N = L$ the zeta values sit consistently *below* CUE: $r_\zeta - r_{\text{CUE}}(N=L) = -0.019$, essentially flat across all windows. Thus ζ is bracketed by the two nulls,

$$\underbrace{0.505}_{\text{Gaussian, same spectrum}} \ll \underbrace{r_\zeta \approx 0.82}_{(\text{at } L \approx 8.4)} < \underbrace{0.841}_{\text{CUE, } N=L},$$

strongly coupled like a random-matrix model, but measurably less than one.

4. AN EFFECTIVE-DIMENSION ANOMALY

To express the deficit structurally, define $N_{\text{eff}}(L)$ by $r_{\text{CUE}}(N_{\text{eff}}) = r_{\zeta}(L)$, inverting a smooth fit $r_{\text{CUE}}(N) = r_{\infty} + c_1/N + c_2/N^2$ of Table 2 (fit RMS 0.004 Pearson, 0.002 Spearman; the fit is used only as an interpolation device — we attach no meaning to its extrapolation). The last two columns of Table 1 give the result.

Observation 1. *Over $5.6 \leq L \leq 12.45$ (heights t from $\sim 1.2 \times 10^3$ to 1.6×10^6),*

$$N_{\text{eff}}^{\text{Pearson}} - L = +0.93, \quad N_{\text{eff}}^{\text{Spearman}} - L = +1.86,$$

each constant in L within errors (constant-model $\chi^2/\text{dof} = 0.09$ and 0.06 ; adding a linear trend improves χ^2 by only 0.5 and 0.1 respectively). We estimate a systematic uncertainty of ± 0.15 on each constant from the interpolation of Table 2.

Observation 2. *No single effective dimension accounts for both statistics: the Pearson and Spearman projections of the joint law $(\tilde{\delta}, \tilde{M})$ require shifts differing by ≈ 0.9 . Consequently the joint gap–maximum law of ζ at finite height is not that of any finite-dimensional CUE.*

Three remarks. First, the constancy of each shift over three orders of magnitude in t makes an accidental origin unlikely; the natural suspects are the prime-oscillation terms of the explicit formula, which modulate $|Z|$ on scales longer than one gap and hence add gap-independent amplitude variance — a decorrelating mechanism that pure CUE lacks. Second, the observable-dependence of N_{eff} is consistent with the picture of [1], where the spacing-shape observable selects the very different effective dimension $N_{\text{eff}} = L/\sqrt{12\Lambda}$, $\Lambda = 1.57314\dots$; each statistic probes the arithmetic corrections through its own projection. Third, we deliberately refrain from proposing closed forms for the constants 0.93 and 1.86 : in an earlier phase of this project two closed-form candidates for a related slope observable survived to $T = 6 \times 10^3$ and $T = 10^4$ respectively and were then cleanly refuted at larger heights; the present note reports only quantities that remained stable under a 20-fold extension in T .

5. OPEN PROBLEMS

(1) Derive the local mean of M_n^2 conditioned on the gap — equivalently the regression slope behind r — from the methods of [2, 3]; the machinery (the auxiliary function Z_1) sums over extrema and may admit a gap-weighted variant. (2) Derive c_{Pearson} and c_{Spearman} , or the deficit $r_{\text{CUE}} - r_{\zeta} = 0.019$, from the explicit formula, in the spirit of [1]. (3) Compute $r_{\text{CUE}}(N)$ analytically; even the $N \rightarrow \infty$ limit (the sine-process value) appears to be unknown. (4) Extend the measurement to Odlyzko-height windows ($t \sim 10^{12}$, $L \approx 26$) to test the constancy of the shifts over a much longer lever arm.

REPRODUCIBILITY

All code (analysis scripts 27–42b), the datasets of gap–maximum pairs, and a dated re-search log are available in the repository <https://github.com/azizran/deney> (directory `qm_riemann/`).

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REFERENCES

- [1] E. Bogomolny, O. Bohigas, P. Leboeuf, A. G. Monastera, *On the spacing distribution of the Riemann zeros: corrections to the asymptotic result*, J. Phys. A: Math. Gen. **39** (2006), 10743–10754. arXiv:math/0602270.
- [2] J. B. Conrey, A. Ghosh, *A mean value theorem for the Riemann zeta-function at its relative extrema on the critical line*, J. London Math. Soc. (2) **32** (1985), 193–202.
- [3] C. Hughes, S. Lugmayer, A. Pearce-Crump, *The second moment of the Riemann zeta function at its local extrema*, arXiv:2411.05573 (2024); J. London Math. Soc. (2025).
- [4] J. P. Keating, N. C. Snaith, *Random matrix theory and $\zeta(1/2 + it)$* , Comm. Math. Phys. **214** (2000), 57–89.
- [5] D. H. Lehmer, *On the roots of the Riemann zeta-function*, Acta Math. **95** (1956), 291–298.
- [6] M. B. Milinovich, *Moments of the Riemann zeta-function at its relative extrema on the critical line*, Bull. London Math. Soc. **43** (2011), 1119–1129.
- [7] H. L. Montgomery, *The pair correlation of zeros of the zeta function*, Proc. Sympos. Pure Math. **24** (1973), 181–193.
- [8] A. M. Odlyzko, *On the distribution of spacings between zeros of the zeta function*, Math. Comp. **48** (1987), 273–308.
- [9] A. Pearce-Crump, *Moments of the Riemann zeta function at its local extrema*, arXiv:2411.05568 (2024); Mathematika **71** (2025), e70035.
- [10] J. Theiler, S. Eubank, A. Longtin, B. Galdrikian, J. D. Farmer, *Testing for nonlinearity in time series: the method of surrogate data*, Physica D **58** (1992), 77–94.

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